

Efficient Exploration of Chemical Compound Space Using Active Learning for Prediction of Thermodynamic Properties of Alkane Molecules

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Cite This: *J. Chem. Inf. Model.* 2023, 63, 6515–6524



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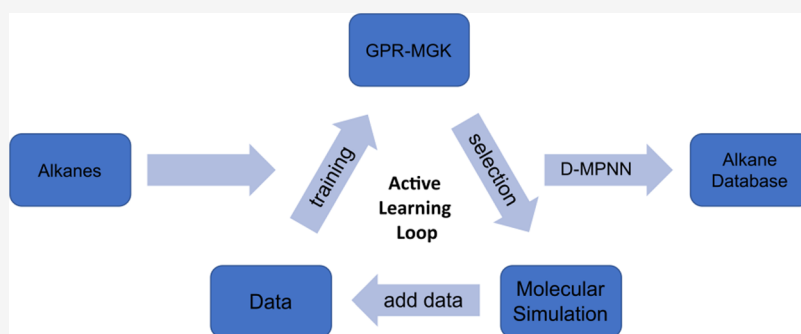
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ABSTRACT: We introduce an exploratory active learning (AL) algorithm using Gaussian process regression and marginalized graph kernel (GPR-MGK) to sample chemical compound space (CCS) at minimal cost. Targeting 251,728 enumerated alkane molecules with 4–19 carbon atoms, we applied the AL algorithm to select a diverse and representative set of molecules and then conducted high-throughput molecular simulations on these selected molecules. To demonstrate the power of the AL algorithm, we built directed message-passing neural networks (D-MPNN) using simulation data as the training set to predict liquid densities, heat capacities, and vaporization enthalpies of the CCS. Validations show that D-MPNN models built on the smallest training set considered in this work, which consists of 313 molecules or 0.124% of the original CCS, predict the properties with $R^2 > 0.99$ against the computational data and $R^2 > 0.94$ against the experimental data. The advantage of the presented AL algorithm is that the predicted uncertainty of GPR depends on only the molecular structures, which renders it compatible with high-throughput data generation.

INTRODUCTION

Active learning (AL) algorithms attempt to select the most valuable data points for labeling, maximizing the performance of the machine learning (ML) model at a minimal cost.^{1–3} AL procedure is usually done by iteration starting from a small subset of the data. At each iteration, a surrogate ML model is trained using the subset to select more data that are valuable for the training. The selection preference depends on the acquisition function of the model. Generally speaking, if the acquisition function is *predicted uncertainty*, the AL algorithm prefers to select samples dissimilar to the training set, a.k.a. *uncertainty reduction* or *exploratory AL*; if the acquisition function is *predicted value*, the AL algorithm prefers to select samples with the desired properties, a.k.a. *greedy* or *exploitative AL*.^{4,5} In the context of this article, we focus on exploratory AL for chemical compound space (CCS), for the purpose of building an ML model to predict the thermodynamic properties of the CCS.

Thermodynamic properties are fundamentally important in chemical and biological engineering. Traditionally, they are

obtained through experiments or computational models based on the experimental data. For example, NIST ThermoData Engine (NIST-TDE) provides critically evaluated thermodynamic data of experimental measurements and extended model predictions.⁶ In recent years, molecular simulation (MS) has showcased its potential in calculating thermodynamic properties from first principles.^{7,8} This approach not only provides a safer alternative to experimental measurements but also offers a cost-effective means of data generation.

ML has great potential to revolutionize the chemical information process, expanding the exploration of CCS by orders of magnitude.^{7–14} Due to the rapid development of deep learning and graph neural networks (GNNs) re-

Received: September 6, 2023

Published: October 19, 2023



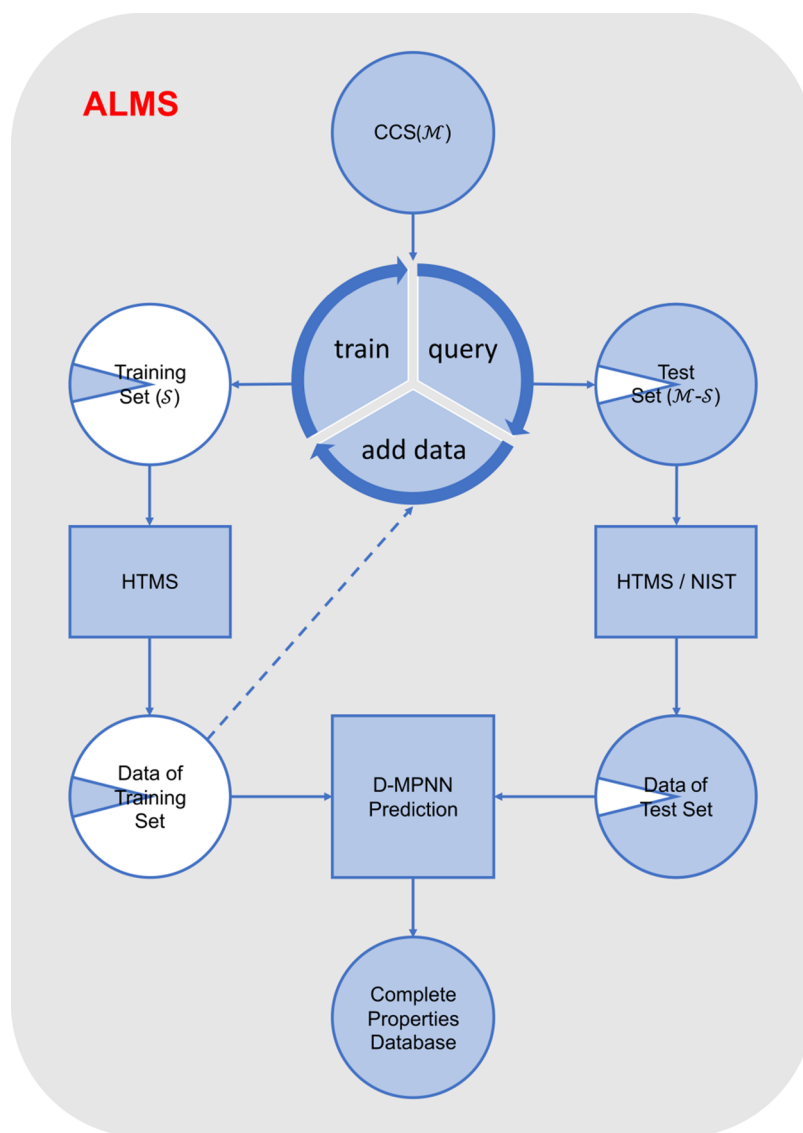


Figure 1. Overview of the active learning molecule simulation (ALMS) workflow. The framework starts with the CCS (M) of interest as input and predictions of their properties as output. An AL algorithm is used to select a training set S from M . The data of S are generated using high-throughput molecular simulation. A D-MPNN is trained to make predictions on the remaining molecules $M - S$.

cently,^{15–18} research on how to use AL to explore CCS has gained renewed attention, especially in drug discovery. For example, Zhang and Lee proposed Bayesian semisupervised graph convolutional neural networks to achieve uncertainty quantification (UQ) and AL;¹⁹ Graff et al. applied batch active learning for high-throughput virtual screening.²⁰ They used directed message-passing neural network (D-MPNN)²¹ as the surrogate model and mean–variance estimation (MVE)²² as the UQ method; Soleimany et al. demonstrated that deep learning outperforms MVE on UQ and AL tasks.²³ However, one of the key challenges to address is the incompatibility between the sequential nature of AL algorithms and the parallel nature of high-throughput simulations or experiments. In fact, most experimental instruments and equipment are designed to test multiple samples simultaneously and not in sequence. Batch AL solves this problem by selecting multiple samples in each AL iteration, but at the cost that the selected samples contain redundant information and the performance is significantly worsened.^{19,20}

In this work, we propose a new exploratory AL algorithm using Gaussian process regression and marginalized graph kernel (GPR-MGK)^{24–26} as the surrogate model. The advantage of this approach is that GPR-MGK uses the molecular graphs (topological structures) to calculate the kernel matrices for similarity comparison and estimation of prediction uncertainty and explores CCS by selecting the most dissimilar molecule which bears the largest prediction uncertainty. Since the ground-truth data are not required for computing the predicted uncertainty, the sample selection cycle for AL can be executed before undertaking expensive data generation. As a result, the proposed AL algorithm is, in principle, compatible with any high-throughput computations or experiments.

We targeted 251,728 enumerated alkane molecules with 4–19 carbon atoms. Varying the predicted uncertainty thresholds (0.5, 0.4, and 0.3) for acquisition, we obtained 313, 599, and 1679 molecules as the training sets, respectively. To test the AL algorithm, we carried out high-throughput molecular simulations (HTMS) with TEAM-AMBER force field^{27,28} to

generate data and trained D-MPNN models for predicting the thermodynamic properties of all molecules in the CCS. The models are validated using computational and experimental testing sets. The computational test set contains 3,000 molecules randomly selected from the CCS excluding those selected by AL and calculated by the HTMS engine. The experimental test set contains 240 molecules with their properties collected from the NIST-TDE database.^{6,29} Although still in its early stages, our study demonstrates the promise of realizing large-scale thermodynamic property predictions through active machine learning and molecular dynamics simulations.

METHODS

Figure 1 illustrates the active learning molecular simulation (ALMS) workflow. The starting point is the CCS $\mathcal{M}$ of interest. A small training set $\mathcal{S}$ is selected using the AL algorithm presented in this work while the remaining molecules $\mathcal{M} - \mathcal{S}$ can be utilized as the test set. The data of $\mathcal{S}$ are generated by HTMS. A D-MPNN is then trained which is able to make accurate predictions on the remaining molecules $\mathcal{M} - \mathcal{S}$ because AL ensures low predicted uncertainty. To validate the ALMS workflow, we constructed a simulated test set and an experimental test set. The simulated test set was also generated using HTMS, while the experimental test set was sourced from the NIST-TDE database. The AL module (hyperparameters of MGK) can be optimized based on the feedback from the generated data, as indicated by the dashed line in the figure. In this work, we found the previously trained GPR-MGK model²⁵ is adequate for selecting the training set. Therefore, we did not further optimize the AL module but instead focused on setting various thresholds for predicted uncertainty.

Exploratory Active Learning. The AL algorithm is shown in Algorithm 1. In the beginning, two molecules are randomly selected from the set $\mathcal{M}$ forming the initial subset $\mathcal{S}$, and the remaining molecules constitute the pool set $\mathcal{P} = \mathcal{M} - \mathcal{S}$. The subset $\mathcal{S}$ grows by using the AL algorithm to select molecules from $\mathcal{P}$. The AL algorithm uses GPR-MGK as the surrogate model. At each iteration, the model is trained using $\mathcal{S}$ to predict the uncertainties of $\mathcal{P}$, and a molecule α with the highest predicted uncertainty is moved from $\mathcal{P}$ to $\mathcal{S}$. The uncertainties of $\mathcal{P}$ are calculated using

$$\mathbf{U}_p = \text{diag}(\mathbf{K}_{pp} - \mathbf{K}_{sp}^T \mathbf{K}_{ss}^{-1} \mathbf{K}_{sp}) \quad (1)$$

where $\mathbf{K}_{pp}$ is the kernel matrix of $\mathcal{P}$, $\mathbf{K}_{ss}$ is the kernel matrix of $\mathcal{S}$, and $\mathbf{K}_{sp}$ is the kernel matrix between $\mathcal{S}$ and $\mathcal{P}$.

During the early stages of AL, calculating the uncertainties for all molecules in $\mathcal{P}$ is prohibitively expensive. To tackle this issue, we implement a solution where, at each iteration, we randomly select N_t molecules from $\mathcal{P}$, unless the number of molecules in $\mathcal{P}$ is fewer than N_t . We then discard molecules that have prediction uncertainties below the threshold u_t while moving the molecule with the highest prediction uncertainty to $\mathcal{S}$. The AL iterations continue until the pool set $\mathcal{P}$ becomes empty, which means all predicted uncertainties have fallen below the threshold value u_t .

The algorithm is an exploratory AL because the criterion for selecting the most valuable molecules is the predicted uncertainty, i.e., the AL algorithm explores the CCS by finding the molecule in $\mathcal{P}$ that is the least similar to $\mathcal{S}$. Therefore, $\mathcal{S}$ is a representative subset of $\mathcal{M}$, and every molecule in $\mathcal{P}$ has

Algorithm 1: exploratory Active Learning using GPR-MGK

Input: $\mathcal{M}$ - target CCS; N_t - number of samples to be examined in each active learning iteration; u_t - uncertainty threshold.

Output: $\mathcal{S}$ - selected set.

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 $\mathcal{S} \leftarrow$  randomly select 2 samples from  $\mathcal{M}$ ;
 $\mathcal{P} \leftarrow \mathcal{M} \setminus \mathcal{S}$ ;
 $\mathcal{C} \leftarrow \emptyset$ : confidence set;
while  $\mathcal{P} \neq \emptyset$  do
  if  $|\mathcal{P}| > N_t$  then
    Randomly select a subset  $\mathcal{T} \in \mathcal{P}$ ,  $|\mathcal{T}| = N_t$ ;
  else
     $\mathcal{T} \leftarrow \mathcal{P}$ ;
  end
   $\mathbf{U}_T \leftarrow \text{diag}(\mathbf{K}_{TT} - \mathbf{K}_{ST}^T \mathbf{K}_{ST}^{-1} \mathbf{K}_{ST})$ : predicted uncertainty;
  if  $\max(\mathbf{U}_T) > u_t$  then
     $\alpha \leftarrow \arg\max(\mathbf{U}_T)$ ;
     $\mathcal{S} \leftarrow \mathcal{S} \cup \{\alpha\}$ ;
     $\mathcal{P} \leftarrow \mathcal{P} \setminus \{\alpha\}$ ;
  end
   $\{\beta_i, i = 1, 2, \dots\} \leftarrow \arg(\mathbf{U}_T < u_t)$ ;
   $\mathcal{C} \leftarrow \mathcal{C} \cup \{\beta_i, i = 1, 2, \dots\}$ ;
   $\mathcal{P} \leftarrow \mathcal{P} \setminus \{\beta_i, i = 1, 2, \dots\}$ ;
end

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corresponding similar molecules in $\mathcal{S}$, which guarantees that $\mathcal{S}$ as the training set can be used to make accurate predictions for all molecules in $\mathcal{P}$. Furthermore, the molecules in $\mathcal{S}$ are diversified, which ensures the number of molecules in $\mathcal{S}$ is minimized and so is the cost of data generation.

The MGK is used to compute the kernel matrix describing the similarity between molecules by comparing random walking paths on a pair of molecular graphs. We applied the MGK proposed in our previous work,²⁵ which is designed to predict the thermodynamic properties of molecular liquids. The formula of MGK is expressed as

$$K(G, G') = \sum_{l=1}^{\infty} \sum_h \sum_{h'} \left[p_s(h_1) p'_s(h'_1) \kappa_v(v_{h_1}, v'_{h'_1}) p_q(h_l) p'_q(h'_l) \right. \\ \times \left(\prod_{i=2}^{l-1} p_t(h_i | h_{i-1}) \right) \left(\prod_{i=2}^{l'-1} p'_t(h'_i | h'_{i-1}) \right) \\ \times \left. \left(\prod_{i=2}^{k-1} \kappa_v(v_{h_i}, v'_{h'_i}) \kappa_e(e_{h_i h_{i-1}}, e'_{h'_i h'_{i-1}}) \right) \right] \quad (2)$$

where G and G' are the two molecular graphs to be computed; $\mathbf{h}$ and $\mathbf{h}'$ are the random walk paths of length l ; p_s , p_q , and p_t are the starting probability, stopping probability, and transition probability, of the random walk process, respectively. The atom kernel $\kappa_v(\bullet, \bullet)$ computes the similarity between a pair of atoms as the product of atomic feature kernels

$$\kappa_v(v, v') = \prod_j \delta_j(\phi_j(v), \phi_j(v')) \quad (3)$$

where the atomic feature kernel $\delta_j(\bullet, \bullet)$ computes the similarity between a pair of atomic features, usually using a Kronecker delta function

$$\delta(\phi_1, \phi_2) = \begin{cases} 1 & \text{if } \phi_1 = \phi_2 \\ h \in (0, 1) & \text{otherwise} \end{cases} \quad (4)$$

The bond kernel $\kappa_e(\bullet, \bullet)$ is defined in the same way as the atom kernel. In addition, the MGK is normalized to improve prediction performance

$$\bar{K}(G, G') = \frac{K(G, G')}{\sqrt{K(G, G)K(G', G')}} \times \exp\left[-\frac{(K(G, G) - K(G', G'))^2}{\lambda^2}\right] \quad (5)$$

The atom features, bond features, and hyperparameters were tuned by minimizing the cross-validation error using the critical temperature data set. We demonstrated that the model offers high-quality uncertainty quantification.^{25,26} The quality of the predicted uncertainty is critical because it guarantees that the selected samples are representative and diversified.

In our previous publication, we reported that the MGK hyperparameters optimized using the critical temperatures data set work well for predicting other properties such as density.²⁵ The transferability of the MGK hyperparameters enables its application to the AL of this work. This approach is rooted in the concept of transfer learning,³⁰ where the hyperparameters represent “knowledge” learned from one data set that can be effectively transferred to other data sets. It is important to note that the prediction uncertainty of the GPR, as defined in eq 1, is solely dependent on the kernel matrices. When the hyperparameters are fixed, the prediction uncertainty depends exclusively on the molecular structures. This characteristic renders the AL algorithm independent of target molecular properties. Consequently, the AL algorithm is exceptionally well-suited for high-throughput data generation tools, whether they are experimental or computational.

High-Throughput Molecular Simulation. We employed the HTMS workflow and TEAM-AMBER force field developed by Gong et al.^{27,28,31} to generate the liquid density, heat capacity, and heat of vaporization (HOV) of alkane liquids. Liquid density was computed by averaging the densities of the trajectories. Heat capacity was computed by combining the intramolecular and intermolecular contributions

$$C_p = \underbrace{C_{\text{trans}} + C_{\text{rot}} + C_{\text{vib}}}_{\text{intra}} + \underbrace{\left(\frac{dU_{\text{inter}}}{dT}\right)_p + \left(\frac{PdV}{dT}\right)_p}_{\text{inter}} \quad (6)$$

where C_{trans} and C_{rot} are the ideal translational and rotational contributions computed through the equipartition theorem, C_{vib} is the vibrational contribution computed using the hindered-rotor model,³² and U_{inter} is the intermolecular potential. In the literature,^{33,34} HOV is commonly computed as

$$H_{\text{vap}} = RT - U_{l,\text{inter}} + C \quad (7)$$

$$C = U_{g,\text{intra}} - U_{l,\text{intra}} + U_{g,\text{inter}} + PV_g - RT - PV_l \quad (8)$$

where $U_{l,\text{inter}}$, $U_{g,\text{inter}}$, $U_{l,\text{intra}}$, and $U_{g,\text{intra}}$ are intermolecular and intramolecular potential of liquid and gas, respectively. Ignoring eq 8 will systematically overestimate HOV, and Gong et al. proposed an empirical approximation for alkanes

$$C_e = -\frac{1}{15}n_C RT \quad (9)$$

where n_C is the number of carbon atoms, and the coefficient $\frac{1}{15}$ was obtained by fitting the experimental data.²⁸

For each molecule, the simulation temperatures were set to 16 points ranging from $0.4 T_c$ to $0.9 T_c$, and the simulation pressure was set to 1 bar, where the critical temperature T_c was predicted by a D-MPNN²¹ trained on experimental data from the NIST-TDE database.^{6,29}

The HTMS takes the SMILES strings identifying molecules, temperature points, and pressure as inputs, and it automatically generates all required input files, submits the jobs to high-performance computer clusters, analyzes the simulation trajectory, and collects the simulation results. Several program packages are used in the workflow. OpenBabel³⁵ and Packmol³⁶ are called to build liquid simulation models. The Gaussian 16 is used for quantum chemistry calculations.³⁷ GROMACS is used for molecular dynamics simulations.³⁸ The workflow is monitored by quality control tests that perform the following tasks: (1) checking if the distribution of kinetic energy satisfies the Maxwell–Boltzmann distribution;³⁹ (2) checking if the density is in a reasonable range; (3) checking if the diffusion coefficient is in a reasonable range; and (4) checking if the temperature, pressure, density, and potential energy are converged.⁴⁰ For details on the HTMS workflow, we refer the reader to ref 28.

Property Prediction Using D-MPNN. The liquid density, heat capacity, and HOV at a given temperature and fixed pressure of 1 bar were calculated and collected as a training data set $\{(x_i, T_i, \rho_i, C_{p,i}, H_{\text{vap},i}), i = 1, 2, 3, \dots\}$ using HTMS, where $x_i \in \mathcal{S}$ is molecule, T_i is temperature, ρ_i is density, $C_{p,i}$ is heat capacity, and $H_{\text{vap},i}$ is HOV. The data set is used to train three D-MPNN models which take molecules and temperatures as input and density, heat capacity, and HOV as outputs, respectively.

D-MPNN²¹ is a variant of message-passing neural network (MPNN)¹⁶ which predicts molecular properties with molecular graphs and RDKit features⁴¹ as inputs. D-MPNN contains two phases: message-passing and readout. In the message-passing phase, messages that are fixed-length vectors of embedding atoms (nodes) and bonds (edges) are passed through bonds, and the hidden states of atoms and bonds are updated according to the incoming messages. The hidden states of atoms are aggregated by the *mean* function to obtain a feature vector. The molecular representation is the concatenation between the feature vector and the RDKit feature vector. The readout phase is a fully connected neural network that connects the molecular representation and target property. We applied the Chemprop library⁴² for training and prediction of D-MPNN with default settings except for the followings: 100 epochs for the training; the temperature is incorporated by concatenating the molecular representation with temperature between the message-passing and readout phases.

We chose D-MPNN because its computational cost is lower than GPR-MGK. Our previous benchmark study shows that D-MPNN predictions exhibit a strong correlation with those of GPR-MGK at a per-sample level.²⁶ Therefore, it is reliable to initially employ GPR-MGK AL for sample selection in constructing the training set followed by using D-MPNN for predictions.

Molecular Enumeration. We take alkane molecules with 4–19 heavy atoms as the target CCS. The molecules are identified by SMILES strings and created by an atom-addition algorithm. Alkane molecules with N heavy atoms are generated

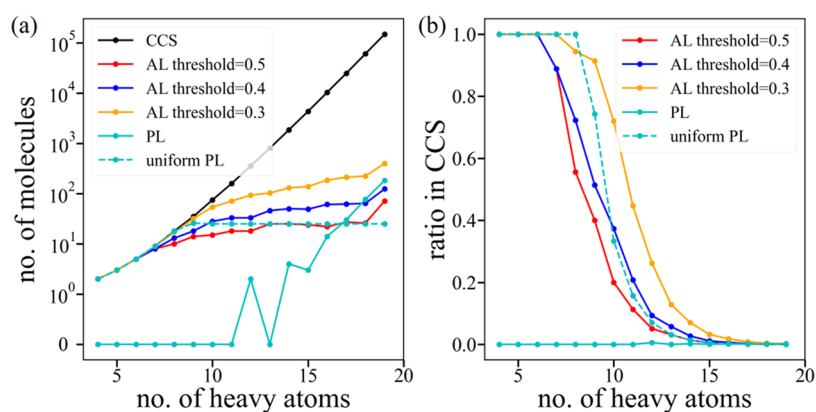


Figure 2. (a) Relationship between the number of molecules and the number of heavy atoms for the CCS, AL sets, and PL sets. (b) Relationship between the ratio of the number of molecules in the AL and PL sets to the CCS and the number of heavy atoms.

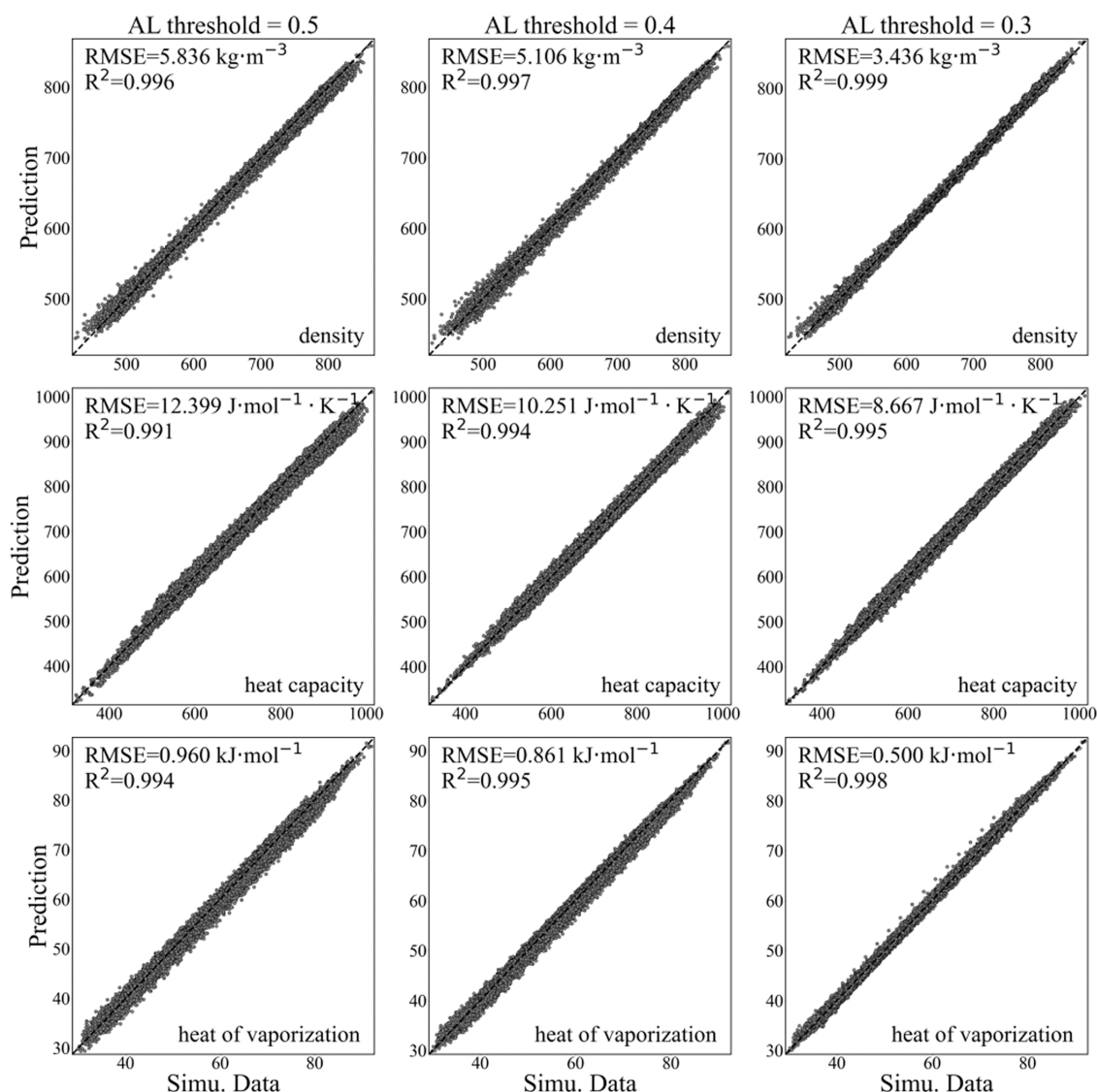


Figure 3. Comparison of D-MPNN predictions to simulation data. The columns show that the training sets are selected through active learning with thresholds 0.5, 0.4, and 0.3. The rows show the target properties: density, heat capacity, and heat of vaporization.

by adding one carbon atom to the alkane molecules with N-1 heavy atoms. The duplicate SMILES strings were discarded by the canonicalization function in RDKit.⁴³ In total, we collected 251,728 distinct molecules.

RESULTS AND DISCUSSION

Heavy Atom Distribution of AL-Selected Molecules.

From the 251,728 target molecules, the training sets were selected using the AL model depending on the uncertainty

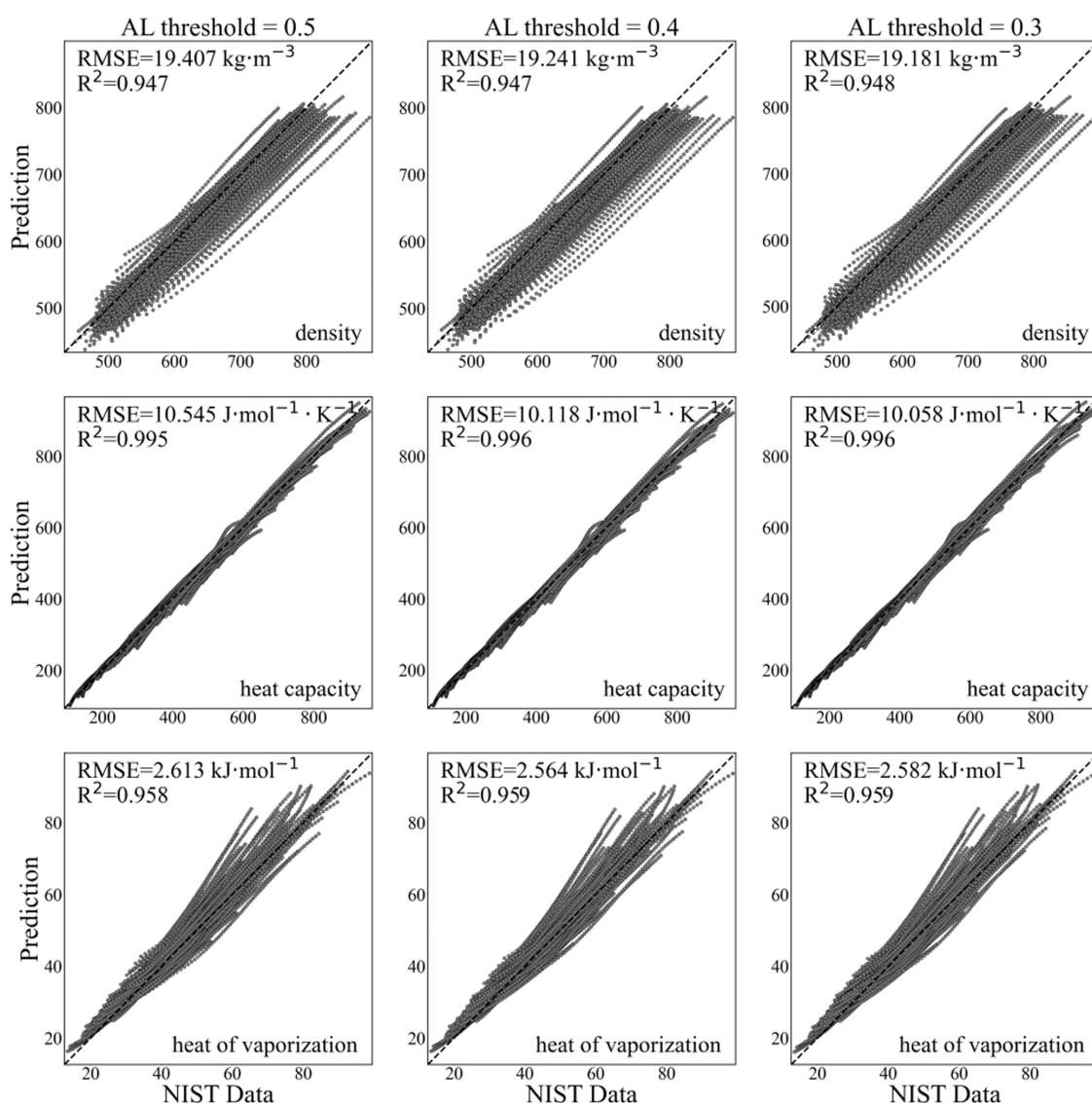


Figure 4. Comparison of ALMS predictions with experimental (NIST-TDE) data. The three columns indicate different training sets obtained by active learning with thresholds of 0.5, 0.4, and 0.3, respectively. The three rows represent the three target properties: density, heat capacity, and heat of vaporization.

thresholds. Using $u_t = 0.5$, 0.4, and 0.3, the training sets are $S_{u_t=0.5}$ (313 molecules), $S_{u_t=0.4}$ (599 molecules), and $S_{u_t=0.3}$ (1679 molecules). Starting from the largest threshold, we collected the first training set and then added more molecules by reducing the threshold to obtain the next training set. Therefore, the training set with a high uncertainty threshold is a subset of the training set with a low uncertainty threshold: $S_{u_t=0.5} \in S_{u_t=0.4} \in S_{u_t=0.3}$.

The distribution of the number of molecules with the number of heavy atoms in the CCS and training sets is shown in Figure 2a. The number of molecules in the CCS grows exponentially with the number of heavy atoms, while the AL-selected training sets grow considerably slower. This suggests that the distribution of molecules in CCS becomes denser, and the proportion of molecules selected by AL becomes less as the number of heavy atoms increases, as indicated in Figure 2b. This phenomenon is consistent with our chemical intuition: the number of structural variations of alkane molecules with the same number of carbon atoms (i.e., structural isomers) increases exponentially, and many of the isomers are “similar”

to each other. Therefore, it is promising to represent a very large CCS with a minimal subset selected by the AL method. As elucidated below, the ML model that is trained using only a few hundred molecules selected by our AL is able to predict the properties of the entire CCS with high accuracy.

Machine Learning Accuracy—Test Against Simulation Data. To evaluate the prediction accuracy, we first used the simulation data of the testing set S_{test} (3000 molecules, of which 12 molecules failed the quality control tests), which were randomly selected from the molecules not selected by AL. For different training sets and different target properties, the ALMS predictions versus the simulation data of the testing set are compared in Figure 3. For $S_{u_t=0.5}$, the root-mean-square errors (RMSEs) of the predictions are 5.8 kg m⁻³ for density, 12.4 J mol⁻¹ K⁻¹ for heat capacity, and 0.96 kJ mol⁻¹ for HOV. Even though the predictions achieved impressive accuracy with an R^2 value greater than 0.99, further improvements were observed as the uncertainty threshold u_t was decreased to 0.4 and 0.3 (training set increased from 313 to 599 and 1679 molecules).

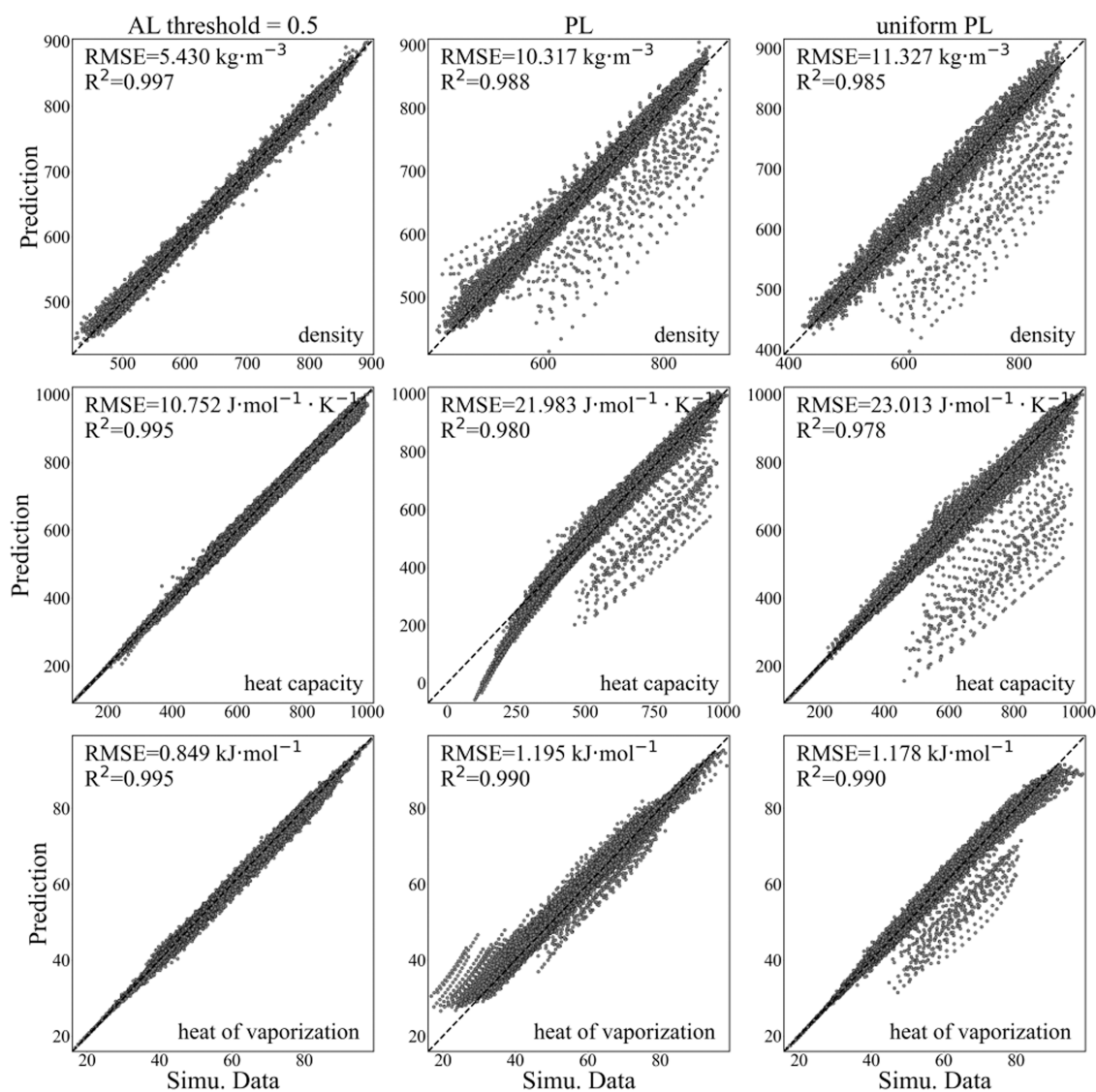


Figure 5. Comparison of ALMS predictions with the simulation data. Left: the training sets are selected through AL with an uncertainty threshold of 0.5. Middle: the training sets of the same size are selected through PL. Right: the training sets of the same size are selected through uniform PL. The rows show the target properties: density, heat capacity, and heat of vaporization.

The uncertainty threshold u_t in the AL serves as a parameter that balances the computational cost and prediction accuracy. Generally, decreasing the value of u_t leads to more simulated molecules and increased computational cost but also results in a more accurate ALMS prediction. For the CCS of alkanes examined in this study, setting u_t to 0.5 was deemed accurate enough for practical applications. Further reduction of u_t requires a significantly greater computational cost, while the improvement in model prediction accuracy is relatively small.

Overall Accuracy—Test Against Experimental Data. We evaluated the ALMS model by comparing the predictions to experimental data. Since our model is built based on simulation data, it brings errors in addition to the ML process. From the NIST-TDE database collected by Knovel,²⁹ we collected experimental data of density, heat capacity, and HOV for 240, 240, and 235 alkane molecules for the purpose. For different AL sets as training sets and different target properties, the predictions of D-MPNNs versus the experimental data of the test set are compared in Figure 4. For $S_{u_t=0.5}$, the RMSE and R^2 are 19.4 $\text{kg} \cdot \text{m}^{-3}$ and 0.947 for density, 10.5 $\text{J} \cdot \text{mol}^{-1} \cdot \text{K}^{-1}$

and 0.995 for heat capacity, and 2.6 $\text{kJ} \cdot \text{mol}^{-1}$ and 0.958 for HOV. The RMSE for density and HOV were found to be twice as high as when the test set was simulated data (Figure 3), indicating that the density and HOV data bear additional errors from the molecular simulations. It is interesting to note, however, that the predictions of the heat capacity agree fairly well with the experimental data. Another important phenomenon is that the prediction accuracy is not sensitive to the size of the training set. The errors of $S_{u_t=0.5}$, $S_{u_t=0.4}$, and $S_{u_t=0.3}$ are almost identical, which indicates that increasing the number of molecules in the training set can only reduce the error of the ML model, not the simulation errors which unfortunately dominate. To improve the quality of our ALMS model, more work is required on the simulation qualities, which are determined not only by the quality of the underlying force field but also by the quality of the data obtained from the simulations.

Active Learning vs Passive Learning. In parallel to the predictions made by using the AL-selected minimal training set, we conducted passive learning (PL) as controlled

experiments to illustrate the advantages of AL. The distribution of molecules in PL sets with the number of heavy atoms is shown in Figure 2. PL refers to random selection of an equal number of molecules (313) within the CCS. And uniform PL refers to PL under the condition that the distribution of heavy atoms number is enforced to be uniform. Note that the random selected test set used in Figure 3 loses its effectiveness because the controlled experiments may pick molecules from it, resulting in a biased comparison. Therefore, we used all simulation data generated in this study as the test set. The predictions of AL, PL, and uniform PL are compared in Figure 5. For the three target properties, the AL prediction is in remarkable agreement with the simulation values, with R^2 scores >0.995 . In contrast, while both PL and uniform PL showcased commendably high R^2 scores ranging between 0.97 and 0.99, a few outliers were still noticeable in the data. The true essence of this advantage stems from the ability to acquire a representative subset that encompasses the entire CCS through AL, which is unattainable with PL or intuitive sampling methods.

As illustrated in Algorithm 1, the AL process commences with the selection of two molecules chosen at random. Consequently, due to the distinct initial molecules, the selected sets produced by the AL also diverge. To establish the robustness of the AL algorithm, we conducted two additional experiments, encompassing AL, PL, and uniform PL methods for molecule selection and subsequent molecular simulations. The results are shown in Figures S1 and S2.

CONCLUSIONS

In this work, we report an exploratory AL algorithm for sampling the CCS efficiently. Coupled with HTMS for data generation and D-MPNN for predictions, we successfully predicted liquid densities, heat capacities, and vaporization enthalpies of all alkane molecules consisting of 4–19 carbon atoms. Remarkably, molecular dynamics simulations were performed on only 313 molecules, which represent a mere 0.124% of the total CCS population, demonstrating the efficiency of the ALMS approach.

The novelty of this work lies in that GPR-MGK is used as the surrogate model of AL. Utilizing GPR-MGK for predicting uncertainty requires only topological structures without the need for ground-truth labels. As a result, AL can be conducted before HTMS. The effectiveness of this approach relies on the GPR-MGK model. We show that a previously developed model, using a critical temperature data set, is transferable and highly effective in predicting various properties presented in this work.

Despite the initial success, several important questions still need to be addressed in the future. For example, in this work, the structure of MGK and the associated hyperparameters are fixed, yielding satisfactory results. However, it remains unclear where the boundary lies for the transferability of the MGK. Hence, it is worth investigating how the hyperparameters of MGK affect the performance of AL, particularly for additional thermodynamic properties. In addition, conducting comparative benchmarks between various AL algorithms using public data sets is essential for gaining a deeper understanding of the relative strengths and weaknesses of the proposed AL algorithm. While molecular simulation holds promise, the need for further enhancements remains apparent. Among the various factors influencing simulation efficiency and accuracy, the quality of the force field stands as a pivotal concern. To

facilitate the development of superior force fields, using either classical functions or machine learning data models, the judicious selection of an optimal training set is imperative.

ASSOCIATED CONTENT

Data Availability Statement

The code for alkane molecules generation is available at <https://github.com/Xiangyan93/molecules-enumerate>. The code for ALMS framework and the simulation data are available at <https://github.com/Xiangyan93/ALMS>. The force field used in this article is available at <http://sun.sjtu.edu.cn/teamff>. The experimental data are taken from NIST database via Knovel at <https://app.knovel.com/web/poc/ms/discovery.html>.

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jcim.3c01430>.

Results of two parallel experiments with ALMS using different random seeds (Figures S1 and S2) (PDF)

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Author Contributions

Y.X., Y.T., H.L., and Z.G. wrote the codes; Y.X. conducted the experiments and analyzed the data; Y.X., Y.T., G.L., and H.S. wrote the manuscript; Y.T., G.L., L.W., and H.S. supervised the work.

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

The computations in this paper were run on the π 2.0 cluster supported by the Center for High-Performance Computing at Shanghai Jiao Tong University. This work was funded by the

National Natural Science Foundation of China [Grant No. 21973060] and [Grant No. 21673138].

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